

IT Project Management

Topic 4: Cost Planning & Performance

Diploma Level | Module 4.1 - 4.5

4.1 Project Cost Estimates

Importance of Costing

Accurate cost estimates are the foundation of project success. They determine feasibility and secure funding.

- ✓ **Feasibility:** Can we afford to do this project?
- ✓ **Baseline:** Creates a standard to measure performance against.
- ✓ **Funding:** Justifies the investment to stakeholders.

Types of Costs in IT

Understanding cost behavior is crucial for accurate estimation.

- ✓ **Direct Costs:** Salaries, Software Licenses, Servers.
- ✓ **Indirect Costs:** Overhead, Office Rent, Utilities.
- ✓ **Fixed vs. Variable:** One-time hardware purchase vs. monthly cloud usage fees.

4.2 Project Budgeting: Allocation

The process of aggregating estimated costs to establish an authorized cost baseline.



1. Allocate Total

Distribute the total project estimate to individual work packages in the WBS.



2. Aggregate

Sum the costs of work packages to determine the total **Budgeted Cost**.



3. Authorize

Add contingency reserves to form the final **Cost Baseline**.

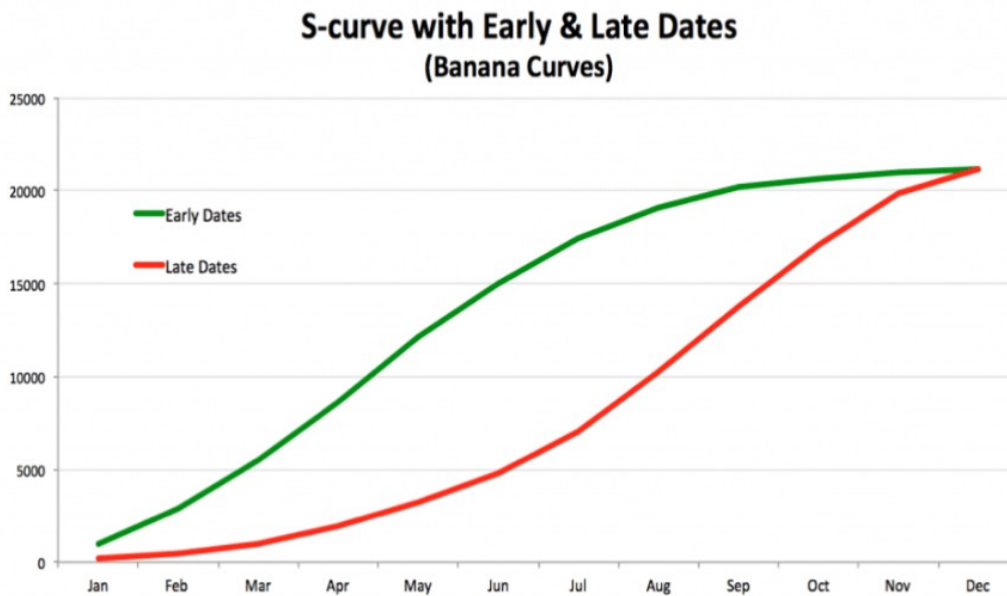
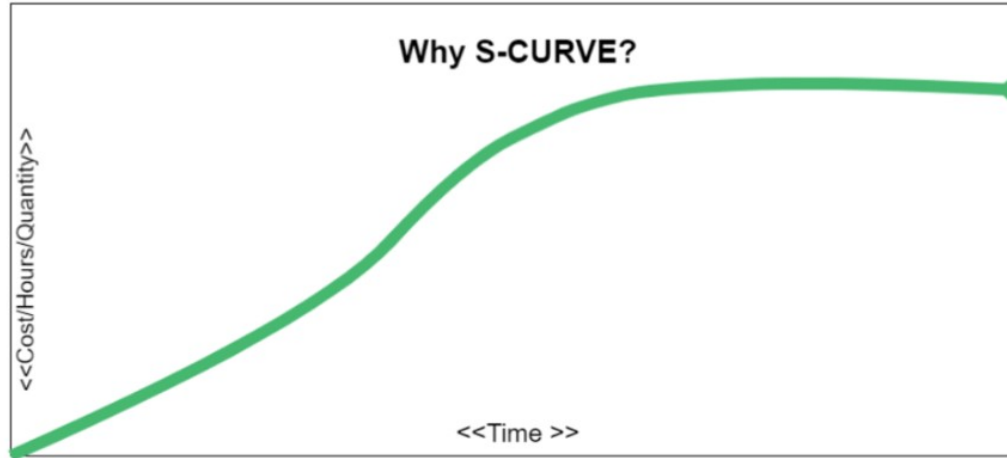
4.2 Cumulative Budgeted Cost

The S-Curve

Project costs are not spent linearly. Budgeting involves developing a time-phased view of costs.

- ✓ **Start:** Low spending (Planning).
- ✓ **Middle:** Peak spending (Execution/Development).
- ✓ **End:** Tapering off (Closure).

This accumulation of costs over time creates the characteristic "S-Curve" used for monitoring performance.



S-Curve

in

Project Management

zerofloat[®]
Σ6LO1109f

4.3 Determining Actual Cost

Actual Cost (AC)

The total cost actually incurred and recorded in accomplishing work performed during a given time period.

- ✓ Salaries paid.
- ✓ Invoices settled.
- ✓ Software subscriptions paid.

Committed Cost

Funds that have been obligated or encumbered but not yet paid. Critical for IT procurement.

- ✓ Signed Purchase Orders (POs) for hardware.
- ✓ Contracts signed with external vendors.
- ✓ **Why important?** Prevents accidental overspending of the "remaining" budget.

4.3 Comparing Actual vs. Budgeted

Regular comparison allows for early detection of cost overruns.

Category	Budgeted Cost	Actual Cost	Variance
Server Hardware	\$10,000	\$12,500	-\$2,500 (Over)
Software Licenses	\$5,000	\$4,800	+\$200 (Under)
Developer Hours	\$20,000	\$18,000	+\$2,000 (Under)
Total	\$35,000	\$35,300	-\$300 (Over)

4.4 Determining Value (EV)

50%

Budget Spent

Just because you spent 50% of the money,
doesn't mean you are 50% done.

EV

Earned Value

The estimated value of the work **actually completed**. It measures how well the project is adhering to the plan.

$$\text{EV} = \% \text{ Complete} \times \text{Budget at Completion (BAC)}$$

4.5 Cost Performance Analysis

Cost Variance (CV)

The amount of budget deficit or surplus at a given point in time.

$$CV = EV - AC$$

Result: Negative = Over Budget.

Cost Performance Index (CPI)

A measure of the cost efficiency of budgeted resources.

$$CPI = \frac{EV}{AC}$$

Result: < 1.0 = Cost Overrun.

4.5 Managing Cash Flow

Liquidity is King

Cost performance is about budget; Cash flow is about having money in the bank to pay bills.

- ✓ **Inflow:** Milestones payments from client.
- ✓ **Outflow:** Monthly salaries, hardware purchases.
- ✓ **The Gap:** IT projects often have large upfront outflows (hardware) before inflows arrive.

Project Managers must align the **Project Schedule** with the **Funding Limit Reconciliation**.

COMPANY NAME	START DATE	END DATE	DATE OF LAST UPDATE

[illegible][illegible][illegible][illegible]

Topic 4 Summary



Budget

Create the Cost Baseline (S-Curve) by aggregating estimates.



Track

Monitor Actual Costs (AC) and Committed Costs.



Analyze

Use CPI and CV to determine if the project is adhering to the budget.

Questions?

IT Project Management: Cost Planning &
Performance

Image Sources



<https://www.zerofloat.com.au/wp-content/uploads/2021/07/SCurve.png>

Source: www.zerofloat.com.au



<https://www.smartsheet.com/sites/default/files/2020-09/IC-Project-Cash-Flow-Template.png>

Source: www.smartsheet.com